



STRATHMORE UNIVERSITY
Strathmore Institute of Mathematical Sciences
MSc in Data Science and Analytics

Sh73 Billion Collected, 1,795 Homes Built: Chaining Vulnerability, Efficiency, and Allocation in Kenya's Affordable Housing Programme

Dissertation Proposal

Student: Valerie Jerono
Registration Number: 222331
Supervisor: Dr. Titus Orwa

A proposal submitted in partial fulfilment of the requirements
for the degree of Master of Science in Data Science and Analytics
at Strathmore University

August 2026

Abstract

Kenya’s Affordable Housing Levy has, in its first full year of national collection, mobilised KSh 73.2 billion against a Treasury target of KSh 63.2 billion , yet the same reporting period closed with only 1,795 completed housing units nationally, against an official annual delivery target of 200,000 units, and with more than KSh 30 billion of already-collected levy proceeds sitting undisbursed (Business Daily Africa, 2025). This gap is the problem this proposal exists to solve: fiscal capacity is not the binding constraint on Kenya’s housing programme, yet county-level allocation of the Levy fund and unit quota is currently decided through fragmented, manual, tabular reviews that cannot distinguish a county’s genuine housing-delivery capacity from its revenue entitlement, its population share, or its historical precedent, and that offer no mechanism for tracing a shilling of allocated funding back to the household-level risk evidence that should justify it. We propose to design, validate, and deploy a single, chained CRISP-DM decision-support pipeline that will close this gap in three linked stages: (i) a leakage-audited, spatially cross-validated machine-learning model that will score household-level housing financial vulnerability from the 2023/24 Kenya Housing Survey; (ii) a non-radial Data Envelopment Analysis layer that will benchmark each of Kenya’s 47 counties on its efficiency at converting housing-delivery inputs into adequate-housing outputs, operationalising “delivery capacity” as a measurable, county-specific quantity rather than an assumption; and (iii) a Mixed-Integer Linear Programme that will allocate the statutory Levy budget and unit quota across counties under an explicit, county-specific delivery-capacity constraint, separate from the budget constraint, so that the pipeline will be able to establish , for the first time, county by county , whether it is revenue or delivery capacity that actually binds the programme’s impact. The framework will be assessed on predictive validity (Tier 1), bootstrap-validated efficiency estimates (Tier 2), and constraint-binding diagnostics and optimality gap under competing equity regimes (Tier 3), with an independent qualitative case study serving as an external check on the resulting county rankings. We expect the pipeline to produce an auditable, reproducible, policy-legible decision-support artefact , usable directly by the State Department for Housing and Urban Development and the Affordable Housing Board , capable of showing which constraint, revenue or capacity, is actually limiting a given county’s housing outcomes, and what it would take to relax it.

Keywords: housing delivery capacity, household financial vulnerability, machine learning, data envelopment analysis, mixed-integer programming, county resource allocation, Kenya Housing Survey, Affordable Housing Levy

1. Introduction

1.1. Background

Housing sits at the intersection of welfare policy, macroeconomic planning, and constitutional obligation, a framing codified globally under Sustainable Development Goal 11 and restated continentally under African Union Agenda 2063, Aspiration 1 (AMG Advocates, 2025). Kenya’s 2010 Constitution operationalises the same guarantee domestically through Article 43(1)(b), and Kenya Vision 2030 designates affordable housing a flagship pillar of the Bottom-Up Economic Transformation Agenda (BETA) , placing this proposal inside an active, funded national policy priority rather than a peripheral academic exercise (Republic of Kenya, 2024).

That priority is now backed by real money, and the money is not the problem. In the 2024/25 financial year, the Kenya Revenue Authority collected KSh 73.2 billion in Housing Levy receipts,

a 35% increase on the year before and 115.8% of the National Treasury’s own target, with total cumulative Levy collections since 2024 now running past KSh 170 billion (Business Daily Africa, 2025). Set against this, national delivery in the same review year came to just 1,795 completed units , fewer than one percent of the 200,000-unit annual goal , while a Treasury report tabled in Parliament confirmed that more than KSh 30 billion of already-collected levy proceeds remained unspent in Treasury bills rather than converted into housing (Business Daily Africa, 2025). Read together, these two facts describe a programme whose revenue side is working roughly as designed and whose delivery side is not, and they reframe the central housing-policy question away from “how much money is available” and toward a sharper, county-specific one: *where, precisely, does the constraint actually bind , on the budget, or on the physical capacity to build, permit, and hand over completed units* , and current allocation practice has no mechanism to answer that question for any individual county, let alone all 47 simultaneously.

This is precisely the analytical gap this proposal will close, and the 2023/24 Kenya Housing Survey (KHS) makes closing it possible for the first time. The KHS is the first Kenyan household survey to combine a nationally representative sample of 21,347 households across all 47 counties with variables spanning household, dwelling, land, county-infrastructure, and institutional modules. Read alongside the statutory, ring-fenced Levy fund, it will create the conditions under which a single evidence base can speak to household-level risk, county-level delivery capacity, and fund allocation simultaneously , rather than, as at present, in three disconnected exercises answered by three disconnected offices, none of which is positioned to tell the Cabinet Secretary’s office whether a stalled county needs more money or more capacity.

The regional and methodological literature supplies the individual tools this proposal will combine: survey-derived composite indices have become the default evidence base for allocating scarce social-protection and infrastructure resources where administrative income records are unreliable (Aiken et al., 2022; Razavi et al., 2025), machine learning has been shown to improve materially on proxy-means testing for this purpose when spatial leakage is handled explicitly (Corral et al., 2025; Li et al., 2022), Data Envelopment Analysis (DEA) is the standard non-parametric tool for benchmarking devolved public-service delivery capacity without assuming a production function and has already been applied at exactly Kenya’s 47-county granularity (Achoki et al., 2026), and Mixed-Integer Linear Programming is the natural instrument for turning quantified need and quantified capacity into an executable budget and unit allocation (Dometi, 2025). What none of these literatures does, individually, is chain need, capacity, and allocation together around one country’s one household survey and one statutory fund , which is the specific, defensible contribution this proposal will make.

1.2. Research Problem

County-level allocation of Kenya’s Housing Levy fund and unit quota is, at present, a largely manual, tabular exercise that treats household vulnerability, county delivery capacity, and fund allocation as three unrelated questions answered by three unrelated processes , and the publicly reported outcome of that arrangement is a programme that has comfortably met its revenue target while completing well under one percent of its unit target (Business Daily Africa, 2025). This is not simply inefficient; it is analytically blind in a specific, correctable way. A process built only around revenue and population share cannot distinguish a county that is under-resourced from a county that is resourced but delivery-constrained , by land, permitting, contractor capacity, or infrastructure , and it therefore cannot tell a policymaker where an additional shilling will actually become a housing unit and where it will simply join the KSh 30 billion already sitting idle. A model built only at the household level cannot speak to county capacity constraints; a model built only at the county level cannot identify which households within that county are actually at risk; and neither, on its own, can produce an executable, capacity-aware budget. Absent an integrated framework that will chain

these three layers together , and that will treat delivery capacity as a distinct, measurable, binding constraint rather than an assumption , Kenya’s housing planners will continue to lack a defensible, county-specific answer to the question the levy’s own performance data has already raised: is it money, or is it capacity, that is actually standing between the Housing Levy and a completed home. This proposal is motivated by the absence of exactly that answer, and it will pursue producing it.

1.3. *Research Objectives*

- (i) To develop and validate a leakage-audited, spatially cross-validated machine-learning model of household housing financial vulnerability (HFVS) from the KHS 2023/24 micro-data.
- (ii) To formulate a non-radial, bootstrap-validated Data Envelopment Analysis model of county-level housing-delivery efficiency across Kenya’s 47 counties, incorporating survey-weighted HFVS aggregates as part of its input,output specification, so as to operationalise delivery capacity as a measurable county quantity.
- (iii) To design and solve a multi-objective Mixed-Integer Linear Programme that allocates the statutory Affordable Housing Levy fund and unit quota across counties under an explicit, county-specific delivery-capacity constraint held distinct from the budget constraint, in order to establish, county by county and under contrasting equity regimes, whether revenue or delivery capacity is the binding constraint on programme impact.
- (iv) To validate the chained pipeline’s resulting county rankings and constraint diagnoses against an independent qualitative case study, and to assess the framework’s transferability to other East African Community contexts.

1.4. *Significance and Justification*

At the global level, this work will contribute directly to **SDG 1** (No Poverty), **SDG 10** (Reduced Inequalities), and **SDG 11** (Sustainable Cities and Communities, target 11.1 on adequate, safe, and affordable housing). At the continental level, it will respond to **African Union Agenda 2063**, Aspiration 1. At the national level, it will operationalise **Kenya Vision 2030**’s BETA housing pillar and will give the **Affordable Housing Act, 2024** a concrete, capacity-aware allocation mechanism, at a moment when the Levy’s own disbursement record , KSh 73.2 billion collected against 1,795 units delivered, and KSh 30 billion left unspent , has already drawn parliamentary and public scrutiny over how, and how well, the fund is being converted into housing (Business Daily Africa, 2025). Practically, the framework will be designed for direct use by the **State Department for Housing and Urban Development**, the **Affordable Housing Board**, the **Insurance Regulatory Authority (IRA)**, and the **Kenya Mortgage Refinance Company (KMRC)**, as a reproducible, versioned decision-support artefact that will close the gap between a household survey and a funded, county-executable housing programme. In doing so, it will offer these stakeholders, and the Cabinet Secretary’s office ultimately accountable for the Levy’s disbursement, a defensible, re-runnable answer to the question their own performance data has raised but their current processes cannot answer: precisely which counties are capacity-constrained rather than revenue-constrained, what it would cost to relax that constraint, and which households the next tranche of funding should be shown to reach.

2. Literature Review

This review moves from broad to narrow across four literatures , household-level vulnerability prediction, spatial validity in survey-based machine learning, county-level efficiency benchmarking (within which the anchor study for this proposal’s Tier 2 design is identified and discussed in detail), and optimisation-based resource allocation , before synthesising the scientific gap that motivates the chained design proposed here.

2.1. Household-level vulnerability and wealth prediction

A first, broad stream of work has shown that supervised machine learning, applied to household survey micro-data, outperformed traditional linear proxy-means testing for identifying at-risk households in low- and middle-income settings. Aiken et al. (2022) combined mobile-phone metadata with humanitarian programme records across several countries; their gradient-boosted classifiers substantially outperformed geographic or categorical eligibility rules for identifying the poorest households, establishing both the problem framing this proposal adopts (cheap, reliable identification of at-risk households) and its core method (tree-ensemble classification with post-hoc explainability). Corral et al. (2025) extended comparable methods to poverty-mapping tasks on nationally representative survey data and found that machine learning trained on remotely sensed covariates underperforms relative to traditional survey-based estimation primarily because of the low predictive capacity of those remotely sensed covariates rather than measurement error alone; Li et al. (2022) confront a related measurement constraint, in which household income or consumption is frequently recorded in coarse brackets that cap achievable R^2 regardless of model sophistication and push practitioners toward classification or ranking framings, a measurement constraint this proposal will confront directly by engineering a continuous cost-burden ratio rather than relying on raw income bands. Muñetón-Santa and Manrique-Ruiz (2023) applied a comparable open-data, spatial machine-learning approach to multidimensional poverty prediction and found that ensemble tree methods consistently outperformed logistic baselines once spatial covariates were included, but that these gains evaporated under naive random-split validation, the same warning that Kapoor and Narayanan (2023) and Wang et al. (2023) generalise below. Razavi et al. (2025) moved from prediction to index construction, using unsupervised PCA over more than 136,000 households across 15 African countries to build a cross-nationally comparable wealth index, validated against macroeconomic benchmarks; this is a direct precedent for the defensible composite vulnerability index this proposal will construct, rather than a single point prediction. Two African infrastructure-specific applications narrow this further: Yitagesu et al. (2025) trained five supervised classifiers on over 500,000 DHS households across 34 Sub-Saharan African countries to predict sanitation access, reporting strong discrimination and using SHAP-based importance ranking to surface housing- and infrastructure-linked predictors; and Akindejoye et al. (2026) constructed a machine-learning composite societal-vulnerability framework for coastal flooding in urban Lagos, combining multiple deprivation domains into a single risk surface, structurally the same composite-index problem this proposal poses for housing financial vulnerability. Mohale and Obagbuwa (2024) closed this cluster by pairing remotely sensed covariates with household survey targets in South Africa and reported that the resulting poverty index supported urban-planning and infrastructure-allocation decisions directly, evidence that a well-validated household index carries demonstrable downstream policy use rather than purely academic interest. On modelling choice, Grinsztajn et al. (2022) benchmarked gradient-boosted trees against deep-learning architectures across 45 tabular datasets and found tree ensembles systematically superior below roughly 50,000 rows, precisely the regime of the KHS household spine, directly motivating this proposal’s choice of gradient boosting over neural alternatives for Tier 1. Aas et al. (2021) examined the reliability of SHAP explanations under correlated tabular features and found that explicit interaction terms improve both accuracy and explanation consistency under moderate collinearity, a finding that will inform how domain-level SHAP weights are constructed for the HFVS composite.

2.2. Spatial validity and leakage in survey-based machine learning

A second, methodological stream addresses a threat that cuts across all of the above: naive random cross-validation has been shown to systematically overstate a model’s ability to generalise to new geographic units. Kapoor and Narayanan (2023) formalised this as a leakage taxonomy across machine-learning-based science broadly and identified spatial and temporal dependence between train

and test folds as among the most common, and most silently damaging, leakage types. Wang et al. (2023) proposed the Spatial+ cross-validation method specifically to address this in geospatial machine learning, splitting on both geographic and feature space rather than geography alone, and found that conventional spatial block cross-validation still under-corrected when feature space was itself spatially structured, precisely the situation in a household survey stratified by county. Together, these two studies justify treating county as a grouping variable for cross-validation rather than as an ordinary categorical feature, and motivate reporting the standard-CV/spatial-CV performance gap as a first-class result rather than a robustness footnote.

2.3. County-level efficiency and delivery-capacity benchmarking with DEA

A third stream evaluates devolved public-service delivery units against one another using DEA, the standard non-parametric technique for benchmarking multiple-input, multiple-output production, in this proposal’s terms, for turning “delivery capacity” from a rhetorical claim into a measured, bounded, county-specific quantity, without assuming a functional form.

Anchor study. Among all reviewed work, Achoki et al. (2026) constitutes this proposal’s single closest precedent and is treated as the anchor study against which the proposed Tier 2 design is directly benchmarked. Their study applied a Simar-and-Wilson bootstrap DEA of technical, pure-technical, and scale efficiency across exactly Kenya’s 47 county governments in the health sector, and subsequently used truncated regression to explain efficiency variation with contextual county covariates. It is the only study identified in this review that operates at the identical administrative unit, Kenya’s 47 devolved county governments, with an identical bootstrap-DEA methodology, applied to a comparable public-service-delivery question. Table 1 summarises how the proposed Tier 2 design will adapt this anchor, with two deliberate departures made explicit: a shift from health-service inputs/outputs to housing-delivery inputs/outputs, and a shift from the anchor’s radial CCR/BCC specification to a non-radial, non-oriented Slack-Based Measure (SBM), so as to expose non-proportional input slacks that a radial score would otherwise mask. Where Achoki et al. (2026) stops at a validated efficiency ranking, this proposal will carry that ranking forward, as the county-specific delivery-capacity constraint, into a statutory funding decision, which is precisely the extension that Section 2.5 identifies as absent from the literature.

Table 1: The anchor study and the proposed departure from it.

	Anchor (Achoki et al., 2026)	Proposed (this study)
Decision-making unit	47 Kenyan county governments	47 Kenyan county governments (unchanged)
Sector	Health-service production	Housing-delivery production
DEA orientation	Radial CCR/BCC	Non-radial, non-oriented SBM
Uncertainty quantification	Simar, Wilson bootstrap CIs	Simar, Wilson bootstrap CIs (retained)
Second-stage use	Truncated regression on covariates	Fed directly into a downstream MILP as a binding delivery-capacity constraint
Endpoint	Validated efficiency ranking	Statutory fund and unit allocation, with a revenue-vs-capacity diagnosis

Beyond the anchor study, Amusa and Fadiran (2024) applied a comparable municipal-efficiency benchmark to 213 South African local and metropolitan municipalities across electricity, water, sewerage, and waste services, confirming that DEA scales well to the large numbers of small-to-medium

administrative units typical of devolved African government structures. Li and Wang (2026) demonstrated a three-stage DEA approach that isolates environmental and contextual noise through a second-stage stochastic-frontier regression before re-scoring efficiency, directly transferable to controlling for county wealth or urbanisation before scoring housing-delivery efficiency. Małyszko et al. (2026) showed how bootstrap 95% confidence intervals around efficiency scores exposed which efficiency differences were statistically meaningful rather than sampling noise, a discipline this proposal will import for a 47-DMU sample where naive point estimates would otherwise over-interpret small efficiency gaps.

2.4. Optimisation-based resource allocation

The final, narrowest stream turns diagnostic scores into an executable allocation. Dometi (2025) reviewed the use of linear- and mixed-integer-programming models within public financial management systems in resource-constrained developing economies and concluded that MILP formulations are well suited to translating equity and efficiency objectives into a budget-and-quota allocation, provided the underlying need and capacity inputs are themselves defensible, exactly the dependency this proposal’s chained design is built to satisfy, and exactly the step at which the existing literature stops short of testing whether a fiscal or a physical constraint actually binds. Legal and policy analyses of the Affordable Housing Act, 2024 (Republic of Kenya, 2024; KPMG East Africa, 2024; AMG Advocates, 2025) establish the concrete statutory parameters, a 1.5% levy on gross income, matched by employers, collected by the Kenya Revenue Authority, and channelled into a ring-fenced Affordable Housing Fund overseen by an Affordable Housing Board, that the MILP’s budget constraint will need to respect, while the Levy’s own published disbursement record (Business Daily Africa, 2025) supplies the empirical motivation for modelling delivery capacity as a second, independent constraint rather than assuming budget is the only one that matters.

2.5. Scientific gap

Read together, these four literatures are individually mature but have not been chained, and none of them treats delivery capacity as a distinct, quantified constraint on allocation. Household-vulnerability studies stop at a risk score; DEA studies, including the anchor study of Section 2.3, stop at a county efficiency ranking without feeding it forward as a hard constraint on any subsequent allocation decision; MILP studies assume need and capacity inputs are already given and typically model only a single budget constraint; and none of the reviewed Kenya- or Africa-specific applications combine all three around a single national household survey and a single, statutorily defined funding instrument whose own public performance record already shows a revenue-delivery disconnect. Several distinct lines of research follow from this synthesis: (a) household-level vulnerability prediction alone, refined toward ordinal or interval targets given coarse income brackets; (b) county efficiency benchmarking alone, extended with dynamic multi-wave DEA once further KHS rounds are released; (c) allocation optimisation alone, treated as a policy-simulation exercise independent of primary data; and (d) the integrated micro-to-allocation-to-validation pipeline, with delivery capacity modelled as a first-class constraint, that this proposal will pursue. We will pursue line (d) as the most defensible, since it is the only one of the four expected to produce a single, end-to-end, auditable artefact capable of telling a real institution, for a real and already-existing statutory fund, whether the constraint it faces in any given county is fiscal or physical.

3. Methodology

We will adopt the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, structured as three chained analytical tiers on a single data spine, followed by an independent validation layer, as illustrated in Figure 1.

Tier 1: Micro ML
Household HFVS

Tier 2: Macro DEA
County Capacity

Tier 3: MILP
Capacity-Aware Levy Allocation

Validation
Case-Study Check

Figure 1: The chained CRISP-DM pipeline: household vulnerability will feed county efficiency and delivery-capacity estimation, which will in turn bound the statutory allocation optimiser, validated externally against an independent case study.

3.1. Data understanding

We will use the Kenya Housing Survey (KHS) 2023/24 as the primary micro-data source: raw KNBS modules covering household, individual, dwelling, land parcels, county physical planning, water services, mortgage, loans, and institutional records, joined on a common household key and county code, yielding a master frame of 21,347 households with full coverage of all 47 counties, together with survey design weights for correct household-to-county aggregation. We will treat two classes of information as exogenous policy inputs, sourced outside the survey: statutory Levy parameters from the Affordable Housing Act, 2024 (the annual Fund order of magnitude, the 1.5% levy rate, and county unit-quota conventions), and county-level delivery-capacity proxies, historical completion counts, project counts, and completion rates by county, drawn from published Affordable Housing Board and National Treasury sector reporting, of the kind that has already documented the 1,795-unit national delivery figure this proposal is motivated by (Business Daily Africa, 2025).

3.2. Data collection

This will be a secondary-data study. KHS 2023/24 was collected by the Kenya National Bureau of Statistics under its existing national sampling design, and no new primary data collection is planned. Statutory levy parameters and county delivery-capacity proxies required for Tier 3 will be sourced directly from the Affordable Housing Act, 2024, its subsidiary regulations, and public Affordable Housing Board / National Treasury sector budget and performance reports.

3.3. Data cleaning

We will apply a documented, staged exclusion and feature-selection process, completeness thresholds, domain-conditional missingness diagnosis, variance-inflation-factor pruning within each HFVS domain, missing-not-at-random-aware imputation, and leakage quarantine of any variable definitionally related to the target, to reduce the raw frame to an audited household-level feature manifest, which we will split 70/15/15 into train/validation/test sets, stratified jointly on county code and urban/rural status. We will construct the vulnerability target as a household cost-burden ratio,

$$\text{HCB}_i = \frac{E_i}{M_i}, \quad (1)$$

where E_i is monthly housing-related expenditure (actual rent for tenants, imputed rent for owners) and M_i is the household's income-bracket midpoint. All definitionally circular expenditure columns will be quarantined out of the predictor set, and all standardisation statistics will be fit on the training partition only and applied, not refit, to any held-out partition, to prevent leakage.

3.4. Data science and optimisation modelling

Tier 1: Household HFVS (micro layer). We will model HCB_i using gradient-boosted tree ensembles, motivated by the tabular-data benchmark of Grinsztajn et al. (2022), under two complementary framings: a continuous regressor, evaluated on test R^2 and mean absolute error, and a top-decile vulnerability classifier ($\text{HCB}_i \geq$ the 90th percentile), evaluated on ROC-AUC and PR-AUC, with isotonic probability calibration and a grid-searched decision threshold p^* chosen to maximise recall

subject to a precision floor. Generalisation will be assessed with both standard k -fold cross-validation and spatial **GroupKFold** cross-validation grouped on county code, following Wang et al. (2023) and Kapoor and Narayanan (2023); we will report the standard-CV/spatial-CV performance gap explicitly, as evidence of whether county fixed effects are required for out-of-county scoring. We will further examine, as a first-class fairness diagnostic rather than an afterthought, whether prediction error is systematically unequal across income quartiles, since a model that under-predicts vulnerability for the poorest households would itself reproduce the allocation bias this proposal is designed to correct. We will construct a five-domain HFVS composite , financial stress, tenure insecurity, physical hazard, structural quality, and utility deprivation , as a SHAP-weighted combination of domain-level sub-scores, informed by the interaction-aware SHAP diagnostics of Aas et al. (2021), and aggregated to county level using survey weights:

$$\text{HFVS}_c = \frac{\sum_{i \in c} w_i \sum_{d=1}^5 \beta_d s_{i,d}}{\sum_{i \in c} w_i}, \quad (2)$$

where $s_{i,d}$ is household i 's standardised score on domain d , β_d is the SHAP-derived domain weight, and w_i is the household's survey weight.

Tier 2: County efficiency and delivery-capacity frontier (macro layer). Following the anchor design of Achoki et al. (2026) (Table 1), we will formulate each of Kenya's 47 counties as a Decision-Making Unit (DMU) and estimate relative housing-delivery efficiency with a non-radial, non-oriented Slack-Based Measure (SBM) DEA model, in preference to the anchor study's radial CCR/BCC specification, since SBM will directly expose non-proportional input slacks:

$$\rho^* = \min \frac{1 - \frac{1}{m} \sum_{r=1}^m \frac{s_r^-}{x_{r0}}}{1 + \frac{1}{n} \sum_{k=1}^n \frac{s_k^+}{y_{k0}}} \quad \text{s.t.} \quad \sum_{c=1}^{47} \lambda_c x_{rc} + s_r^- = x_{r0}, \quad \sum_{c=1}^{47} \lambda_c y_{kc} - s_k^+ = y_{k0}, \quad \lambda, s^-, s^+ \geq 0, \quad (3)$$

with inputs x_r drawn from water-access travel time, overcrowding (persons per room), and an engineered informal-livelihood proxy, and outputs y_k drawn from a dwelling structural-quality index, the share of adequate dwellings, and the county-aggregated financial-stability score $1 - \overline{\text{HFVS}}_c$. We will confirm the DMU-to-variable ratio against Cooper's rule of thumb ($N \geq 3(m + s)$) before finalising the input/output specification, and will compute 95% bootstrap confidence intervals around each county's efficiency score using the Simar-and-Wilson resampling procedure ($B = 300$), following Achoki et al. (2026) and Małyszko et al. (2026), so that county rankings will only be treated as materially different where their intervals do not overlap. Each county's efficiency score ρ_c^* will, in turn, anchor a county-specific delivery-capacity ceiling cap_c , combining the DEA efficiency estimate with the published historical completion-rate proxy for that county, that will be carried forward into Tier 3 as a hard constraint distinct from the fund's budget.

Tier 3: Capacity-aware statutory allocation (allocation layer). We will formulate a Mixed-Integer Linear Programme that allocates the statutory Levy Fund budget B (KSh) and a national unit quota U across counties $c = 1, \dots, 47$, using decision variables $x_c \in \{0, 1\}$ (county activation) and $u_c \in \mathbb{Z}_{\geq 0}$ (units allocated to county c):

$$\max \quad Z = \sum_{c=1}^{47} [\alpha \cdot (1 - \rho_c^*) + (1 - \alpha) \cdot \text{Gap}_c] u_c \quad (4)$$

$$\text{s.t.} \quad \sum_{c=1}^{47} \text{cost}_c u_c \leq B, \quad \sum_{c=1}^{47} u_c \leq U, \quad (5)$$

$$u_c \leq \text{cap}_c, \quad u_{\min} x_c \leq u_c \leq u_{\max} x_c, \quad x_c \in \{0, 1\}, \quad u_c \in \mathbb{Z}_{\geq 0}, \quad (6)$$

where Gap_c is the county’s HFVS-based vulnerability gap, ρ_c^* is its Tier-2 SBM efficiency score, and cap_c is the Tier-2-derived delivery-capacity ceiling described above , so that, unlike the anchor study and the wider MILP literature reviewed in Section 4, the model will hold budget and delivery capacity as two independent, simultaneously binding constraints rather than collapsing them into one. We will solve two policy regimes for comparison , an unconstrained capital-concentration mode and a universal-coverage mode that mandates $x_c = 1, u_c \geq u_{\min} \forall c$, and will, for each regime and each county, record which of the three constraints (budget, national quota, or county delivery capacity) is active at the optimum, so that the pipeline will be able to report, county by county, whether it is revenue or delivery capacity that is actually limiting that county’s allocation. In a second optimisation pass, we will replace the single weighted objective with an Augmented Tchebycheff goal-programming formulation to resolve the adequacy/efficiency/cost-burden trade-off without an arbitrarily fixed α .

3.5. Performance evaluation

Tier 1 will be evaluated on train/validation/test R^2 and MAE (continuous framing) and ROC-AUC, PR-AUC, and calibrated recall at the tuned threshold (classification framing), alongside the standard-CV/spatial-CV gap and the income-quartile error-equity diagnostic described above. Tier 2 will be evaluated on mean SBM efficiency, the count and identity of frontier (fully efficient) counties, and the width of bootstrap confidence intervals. Tier 3 will be evaluated on the branch-and-bound optimality gap (target $< 10^{-4}$), total budget utilisation, unit-allocation coverage, the quantified cost differential between the two policy regimes, and , as the single diagnostic most directly answering this proposal’s research problem , the distribution, across counties and regimes, of which constraint (budget or delivery capacity) binds at the optimum. We will additionally report a cross-tier consistency check , whether high-HFVS, low-efficiency counties receive materially different allocations under the two regimes , as the figure most relevant to a policy reader.

3.6. Deployment

We will package the validated Tier 1,3 artefacts , the preprocessing pipeline, the calibrated classifier and quantile regressors, the SBM efficiency engine, and the capacity-aware MILP solver , as versioned, zero-dependency inference components, and will extend a lightweight prototype interface, currently piloted at <https://housingvulnerabilitydecisiontool.vercel.app/>, into a county-level allocation view with an interactive MILP scenario toggle (Mode A vs. Mode B) and an explicit budget-vs-capacity indicator per county, so that a policy user will be able to inspect, for any county, the chain from household-level evidence through to the KSh amount and unit quota it justifies, and to see immediately whether that county’s ceiling is set by money or by capacity. Figure 2 outlines this deployment path.



Figure 2: Deployment path from validated pipeline artefacts to an interactive, policy-facing decision-support tool that will surface, per county, whether budget or delivery capacity binds.

4. Timeline

Table 2 presents the proposed six-month execution schedule, sequenced so that each tier’s output will become the following tier’s input, with validation running as a continuous check rather than a final-month afterthought.

Table 2: Proposed project timeline.

Phase	Milestones
Month 1	Data understanding and cleaning of the KHS 2023/24 master frame; leakage-quarantine audit; county delivery-capacity proxy sourcing from public AHB/Treasury reports; ethics clearance application submitted.
Month 2	Tier 1 model development: gradient-boosted regressor and classifier, isotonic calibration, standard vs. spatial GroupKFold cross-validation, income-quartile error-equity check, SHAP-weighted HFVS composite.
Month 3	Tier 2 DEA specification and estimation: input,output variable selection, Cooper’s-rule DMU check, SBM efficiency scoring, Simar,Wilson bootstrap confidence intervals, derivation of county delivery-capacity ceilings.
Month 4	Tier 3 MILP formulation and solving: statutory parameter sourcing, capacity-constraint specification, two-regime solve (capital-concentration vs. universal-coverage), binding-constraint diagnostics, Augmented Tchebycheff sensitivity pass.
Month 5	Cross-tier validation; independent qualitative case-study check; dashboard and API deployment.
Month 6	Supervisor review, revisions, and defence preparation.

5. Ethical Considerations

This will be a secondary-data study using de-identified KHS 2023/24 microdata; no new primary data collection involving human subjects is proposed. Because household-level financial and tenure indicators will be used to construct vulnerability scores that could, if published at insufficiently coarse resolution, expose or stigmatise identifiable households or communities, an institutional ethics clearance application (or a documented exemption determination for secondary de-identified data, where applicable under the programme’s research-ethics policy) will be submitted at the outset of the study. County- and DMU-level outputs from all three tiers will be reported with minimum reportable-cell safeguards, and any qualitative case-study component will follow standard informed-consent practice for the respondents involved.

6. Resourcing and Tools

The pipeline will be implemented in Python. Tier 1 will use **scikit-learn**, **LightGBM**, and **XGBoost** for gradient-boosted modelling, chosen on the tabular-data evidence of Grinsztajn et al. (2022), **shap** for domain-weight attribution informed by Aas et al. (2021), and **scikit-learn**’s **GroupKFold** for spatial cross-validation. Tier 2 will use an open-source DEA implementation (e.g. **pyDEA** or an equivalent SBM-capable package) together with a custom Simar,Wilson bootstrap routine. Tier 3 will use **PuLP** or an equivalent MILP solver interface (CBC or, where available, a commercial solver such as Gurobi, to meet the branch-and-bound optimality-gap target). No specialised institutional data-sharing agreement will be required beyond standard KHS microdata access already available through the KNBS research-use channel; statutory levy parameters and delivery-capacity proxies needed for Tier 3 will be drawn from public legislation and public Affordable Housing Board / National Treasury sector reporting. Deployment will reuse and extend the existing lightweight web prototype referenced in Section 3.7 rather than provisioning new institutional infrastructure.

7. Expected Outcomes

This research is expected to produce a leakage-audited, spatially validated household housing-vulnerability model and composite HFVS score for all 47 Kenyan counties, together with a transparent

account of where county-level generalisation is, and is not, reliable, and of whether prediction error is distributed equitably across income quartiles. It is expected to yield a bootstrap-validated county efficiency frontier that will, for the first time, express “delivery capacity” as a measured, county-specific ceiling rather than an assumption, and to identify which counties convert housing-delivery inputs into adequate housing outputs most effectively. It is further expected to produce an executable, statutorily grounded MILP allocation of the Affordable Housing Levy fund and unit quota that will hold budget and delivery capacity as two independently binding constraints, and that will therefore be able to report, county by county and regime by regime, whether revenue or delivery capacity is the constraint actually limiting a given county’s housing outcomes, directly answering the question the Levy’s own published performance record has raised. The overall pipeline is expected to be a validated, deployable decision-support artefact, rather than three disconnected analyses, whose county rankings and constraint diagnoses will have been cross-checked against an independent case study and whose CRISP-DM methodology will be documented clearly enough to be re-calibrated on another East African Community country’s national household survey.

References

- Aas, K., Jullum, M., Løland, A., 2021. Explaining individual predictions when features are dependent: More accurate approximations to Shapley values. *Artificial Intelligence* 298, 103502. doi:[10.1016/j.artint.2021.103502](https://doi.org/10.1016/j.artint.2021.103502).
- Aiken, E., Bellue, S., Karlan, D., Udry, C., Blumenstock, J.E., 2022. Machine learning and phone data can improve targeting of humanitarian aid. *Nature* 603, 864,870. doi:[10.1038/s41586-022-04484-9](https://doi.org/10.1038/s41586-022-04484-9).
- Akindejoye, F. et al., 2026. A machine learning framework for assessing systemic societal vulnerability to coastal flooding. *Journal of Flood Risk Management*. doi:[10.1111/jfr3.70249](https://doi.org/10.1111/jfr3.70249).
- AMG Advocates, 2025. The Affordable Housing Act in Kenya: Key provisions, implications, and opportunities. Legal analysis. URL: <https://www.amgadvocates.com/post/the-affordable-housing-act-in-kenya-a-synopsis>.
- Amusa, H.A., Fadiran, D., 2024. The efficiency of public expenditures on basic services: The case of South African municipalities. *South African Journal of Economics*. doi:[10.1111/saje.12377](https://doi.org/10.1111/saje.12377).
- Achoki, T., Rotich, N., Bett, D.K., Tonui, J., Oanda, T., Lesego, A., Wamai, R., Kinifu, Y., Alam, U., Ochieng, W.O., Were, L.P.O., Schneider, M., 2026. Technical and scale efficiency in health service production in Kenya: Subnational analysis of 47 county governments. *Archives of Public Health*. doi:[10.1186/s13690-026-01969-1](https://doi.org/10.1186/s13690-026-01969-1).
- Business Daily Africa, 2025. Kenya delivers 1,795 affordable homes after collecting Sh73bn. Business Daily, Nairobi. Reporting KRA/National Treasury Housing Levy collection data (FY2024/25) and Affordable Housing Board delivery data tabled before Parliament. URL: <https://www.businessdailyafrica.com/bd/economy/kenya-delivers-1-795-affordable-homes-after-collecting-sh73bn-5268214>.
- Corral, P., Henderson, H., Segovia, S., 2025. Poverty mapping in the age of machine learning. *Journal of Development Economics* 172, 103377. doi:[10.1016/j.jdeveco.2024.103377](https://doi.org/10.1016/j.jdeveco.2024.103377).
- Małyszko, K., Pędziński, B., Maślach, D., Warpechowski, M., Marcinowicz, L., 2026. Crisis-time efficiency in Eastern Poland’s regional hospitals (2015,2024): A data envelopment analysis. *Frontiers in Health Services*. doi:[10.3389/frhs.2026.1715091](https://doi.org/10.3389/frhs.2026.1715091).

- Grinsztajn, L., Oyallon, E., Varoquaux, G., 2022. Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems* 35, 507,520. URL: https://proceedings.neurips.cc/paper_files/paper/2022/hash/0378c7692da36807bdec87ab043cdadc-Abstract-Datasets_and_Benchmarks.html.
- Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* 4, 100804. doi:10.1016/j.patter.2023.100804.
- Republic of Kenya, 2024. The Affordable Housing Act, No. 2 of 2024. Technical Report. National Council for Law Reporting (Kenya Law). URL: <https://new.kenyalaw.org/akn/ke/act/2024/2/eng@2024-03-21/source.pdf>.
- KPMG East Africa, 2024. The Affordable Housing Act, 2024. KPMG Insights. URL: <https://kpmg.com/ke/en/home/insights/2024/03/the-affordable-housing-act-2024.html>.
- Li, Q., Yu, S., Échevin, D., Fan, M., 2022. Is poverty predictable with machine learning? A study of DHS data from Kyrgyzstan. *Socio-Economic Planning Sciences* 81, 101195. doi:10.1016/j.seps.2021.101195.
- Dometi, 2025. The utilization of advanced linear programming models to optimize allocation of limited public resources in the context of public financial management systems. *Kings and Queens Journal of Multidisciplinary Research* 1, 49,60. URL: https://www.researchgate.net/publication/393104626_THE_UTILIZATION_OF_ADVANCED_LINEAR_PROGRAMMING_MODELSTO_OPTIMIZE_ALLOCATION_OF_LIMITED_PUBLIC_RESOURCES_INTHE_CONTEXT_OF_PUBLIC_FINANCIAL_MANAGEMENT_SYSTEMS.
- Mohale, V.Z., Obagbuwa, I.C., 2024. Poverty analysis and prediction in South Africa using remotely sensed data. *Applied Computational Intelligence and Soft Computing* 2024, 5137110. doi:10.1155/2024/5137110.
- Muñetón-Santa, G., Manrique-Ruiz, L.C., 2023. Predicting multidimensional poverty with machine learning algorithms: An open data source approach using spatial data. *Social Sciences* 12, 296. doi:10.3390/socsci12050296.
- Razavi, M., Gaba, C., Crown, W., Nandakumar, A., Ortega, J.A., 2025. Formulation and validation of a regional household wealth index for Sub-Saharan Africa. *PLOS ONE* 20, e0335603. doi:10.1371/journal.pone.0335603.
- Wang, Y., Khodadadzadeh, M., Zurita-Milla, R., 2023. Spatial+: A new cross-validation method to evaluate geospatial machine learning models. *International Journal of Applied Earth Observation and Geoinformation* 121, 103364. doi:10.1016/j.jag.2023.103364.
- Li, Z., Wang, X., 2026. Efficiency evaluation of Chinese Yunnan province county area public service for sports and fitness based on three-stage DEA model. *PLOS ONE* 21, e0340803. doi:10.1371/journal.pone.0340803.
- Yitageasu, G., Alemu, E.A., Worede, E.A., Tigabie, M., Demoze, L., 2025. Machine learning-based prediction of household sanitation facility access in Sub-Saharan Africa: Insights from DHS data (2012,2024). *Archives of Public Health* 83. doi:10.1186/s13690-025-01780-4.